

FINAL EXPLANATION: BIOLOGY GRADE 10

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 10 – SUB-STRAND 3.3 ANIMAL GASEOUS EXCHANGE AND RESPIRATION: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the unit. Think back to the very first lesson, when you measured your own breathing rate and pulse rate at rest and again after 2 minutes of step-ups. Your driving question is: Why does my body work so much harder to get oxygen into my cells during exercise, and what happens inside those cells when there is not enough? Write your answers in the space under each prompt. Use your own Lesson 1 data, your final cumulative model and your Driving Question Board. Write in full sentences and use the scientific words you have learned. Explain the reasons (the 'why' and 'how'), not only what happened. Where a prompt asks you to apply your learning to a new situation, use the same ideas.

Part 1: What Your Body Data Showed

In Lesson 1 you measured your breathing rate and pulse rate at rest and then straight after 2 minutes of vigorous step-ups. (a) Describe what happened to both measurements across the class, using example numbers from the class data table. (b) Which two body systems were responding together? (c) Write down the one observation from that day that your early ideas could not explain.

Write your answer here:

Part 2: Getting Oxygen from the Air into the Blood

Explain how oxygen gets from the air outside your body into your blood during exercise. In your answer: (a) explain how the diaphragm and the intercostal muscles make air move into and out of the lungs, and how this links to your doubled breathing rate; (b) name the four features that make the alveoli an efficient gas exchange surface, and explain what each one does for diffusion; (c) explain why tilapia gills in Lake Victoria need the same four features.

Write your answer here:

Part 3: Transport and the Heart–Lung Link

Trace the journey of one oxygen molecule from an alveolus to a working muscle cell in your leg, naming each structure or process in order. Then explain why your heart rate rose at the same time as your breathing rate, and describe what happens to carbon dioxide on its way out.

Write your answer here:

Part 4: What Happens Inside the Cells

(a) Write the word equation and the balanced chemical equation for aerobic respiration, say where in the cell it happens, and explain how it accounts for the doubling of your breathing rate during step-ups. (b) Explain why your leg muscles burned even though you were breathing hard. Use the terms anaerobic respiration, lactic acid and oxygen debt. (c) Explain why you still breathe heavily for a while after you stop exercising.

Write your answer here:

Part 5: Transfer – Akinyi Swims the Winam Gulf

Akinyi, a competitive swimmer from Kisumu, attempts to cross the Winam Gulf of Lake Victoria. Predict what happens to her breathing rate and pulse rate as she swims hard, and explain why in terms of her muscle cells. Then explain what might cause her muscles to ache if she swims very fast, and what her body does when she gets out of the water. Finally, give one reason why her body temperature would rise.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence from the class data	Accurately uses the Lesson 1 numbers (e.g. 14 to 28 breaths per minute, 72 to over 120 beats per minute) and links them to the explanation throughout.	Mentions that both rates rose, with some correct numbers, but rarely links them to the later explanation.	Gives vague or incorrect data, or no data.
Explaining gas exchange and transport	Explains breathing mechanics with pressure changes, the four alveolar features and the full oxygen pathway including haemoglobin and the heart, with correct diffusion reasoning.	Describes most of the mechanics, features and pathway but with a missing step or limited reasoning.	Lists some structures with little explanation, or has errors in the pathway.
Explaining cellular respiration and lactic acid	Gives both equations correctly, places respiration in the mitochondria, and explains lactic acid, burning legs and oxygen debt as cause and effect.	Gives the equations and a mostly correct explanation of lactic acid with minor gaps.	Equations are incomplete or incorrect, or the burning sensation is not explained.

Transfer to the Akinyi scenario	Applies the whole chain of ideas to the new situation, including oxygen debt and heat, using correct vocabulary.	Applies most ideas to the swimming scenario with some omissions.	Repeats facts without applying them to Akinyi.
Scientific vocabulary and clarity	Uses unit terms accurately and writes in clear, connected sentences.	Uses most terms correctly with generally clear writing.	Uses few scientific terms or uses them incorrectly.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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